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Infrastructure as Code, Git as Truth

A Gentle Intro to GitOps

KN Cloud Adoption Sessions

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Agenda

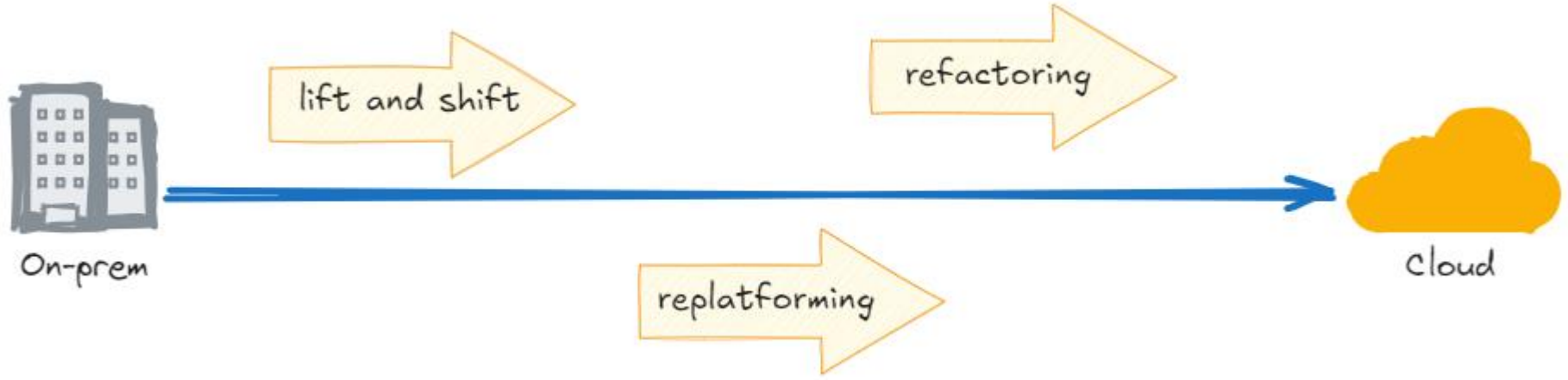
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- Cloud and AWS
- Infrastructure-as-Code (IaC)
- GitOps
- IaC and GitOps
- Strategies
- Use cases
- Q&A



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Cloud Adoption



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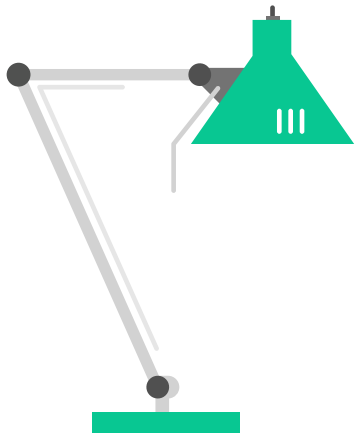
Infrastructure as Code

Infrastructure as Code: Definition

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“Infrastructure as Code (IaC) **is the managing and provisioning of infrastructure through code instead of manual processes**.

With IaC, configuration files are created that contain your infrastructure specifications, which makes it easier to edit and distribute configurations. It also ensures that you provision the same environment every time. By **codifying and documenting your configuration specifications, IaC aids configuration management and helps you avoid undocumented, ad hoc configuration changes**.”



From: What is Infrastructure as Code?, https://www.redhat.com/en/topics/automation/what-is-infrastructure-as-code-iac

Infrastructure as Code: Advantages

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Infrastructure as Code: Strategies

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Imperative

Tell the system **HOW** to do something, detailing every step of the way.



Declarative

Tell the system **WHAT** you want, and let it figure out HOW to accomplish it.

ANSIBLE

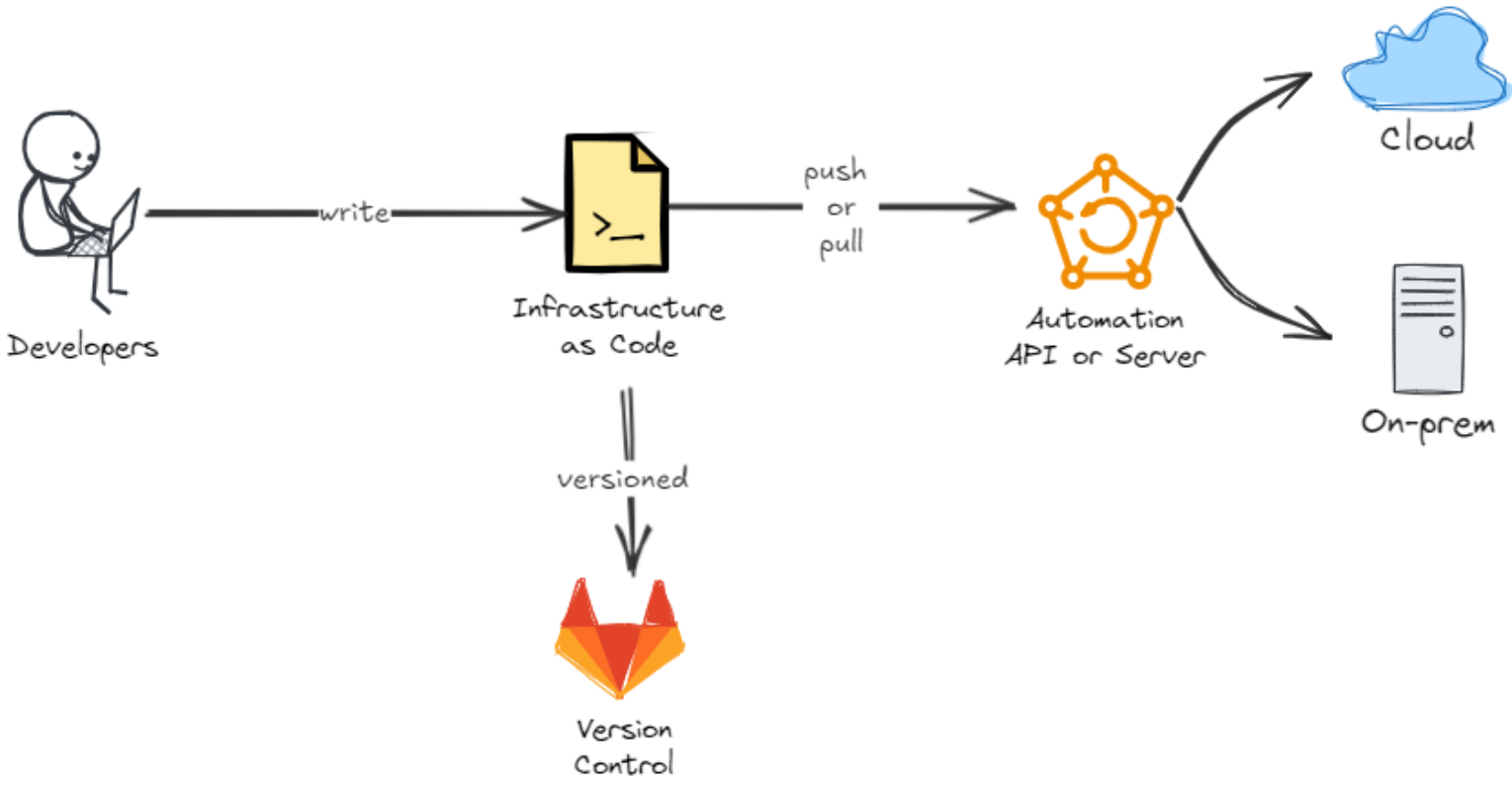
```
#!/bin/bash
```

HashiCorp Terraform

puppet

Infrastructure as Code Workflow

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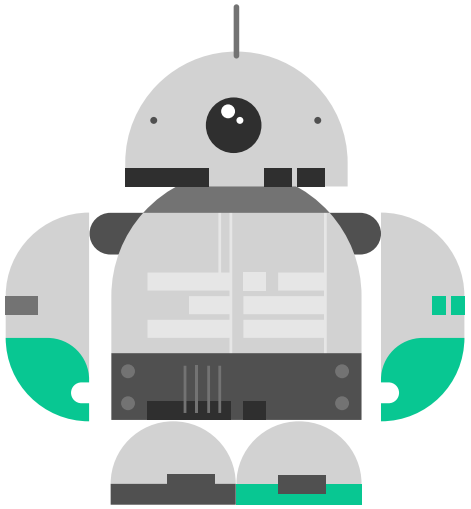
GitOps

GitOps: Definition

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“The core idea of GitOps is having **a Git repository that always contains declarative descriptions of the infrastructure** currently desired in the production environment **and an automated process to make the production environment match the described state in the repository**.

If you want to deploy a new application or update an existing one, you only need to update the repository - the automated process handles everything else. **It’s like having cruise control for managing your applications in production**.”



From: What is GitOps?, https://www.gitops.tech/

GitOps: Advantages

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GitOps: Strategies

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Push-based

Whenever the application code is updated, the build pipeline is triggered, updating both application and infrastructure.



Pull-based

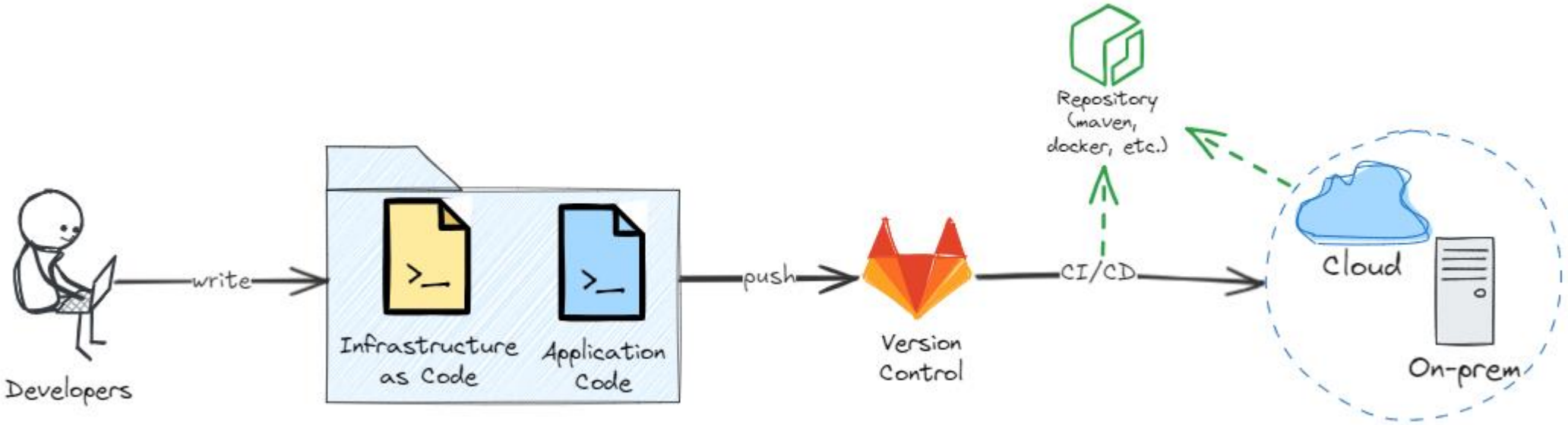
With the pull-based deployment approach, the *operator* is introduced. It takes over the role of the pipeline by continuously comparing the desired state in the environment repository with the actual state in the deployed infrastructure. Whenever differences are noticed, the operator updates the infrastructure to match the environment repository.

Jenkins circleci CI/CD

argo flux

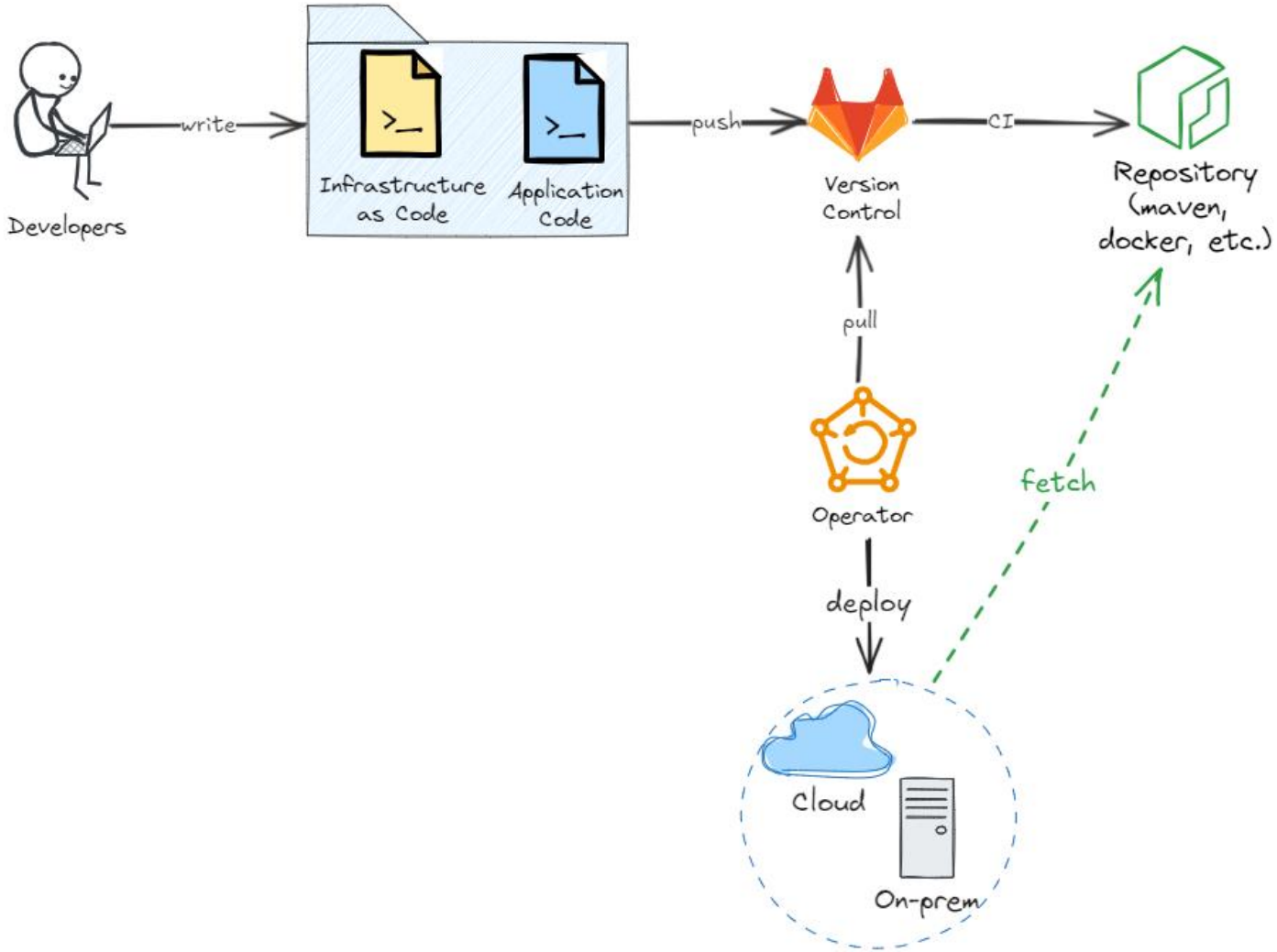
GitOps: Workflow (push-based)

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GitOps: Workflow (pull-based)

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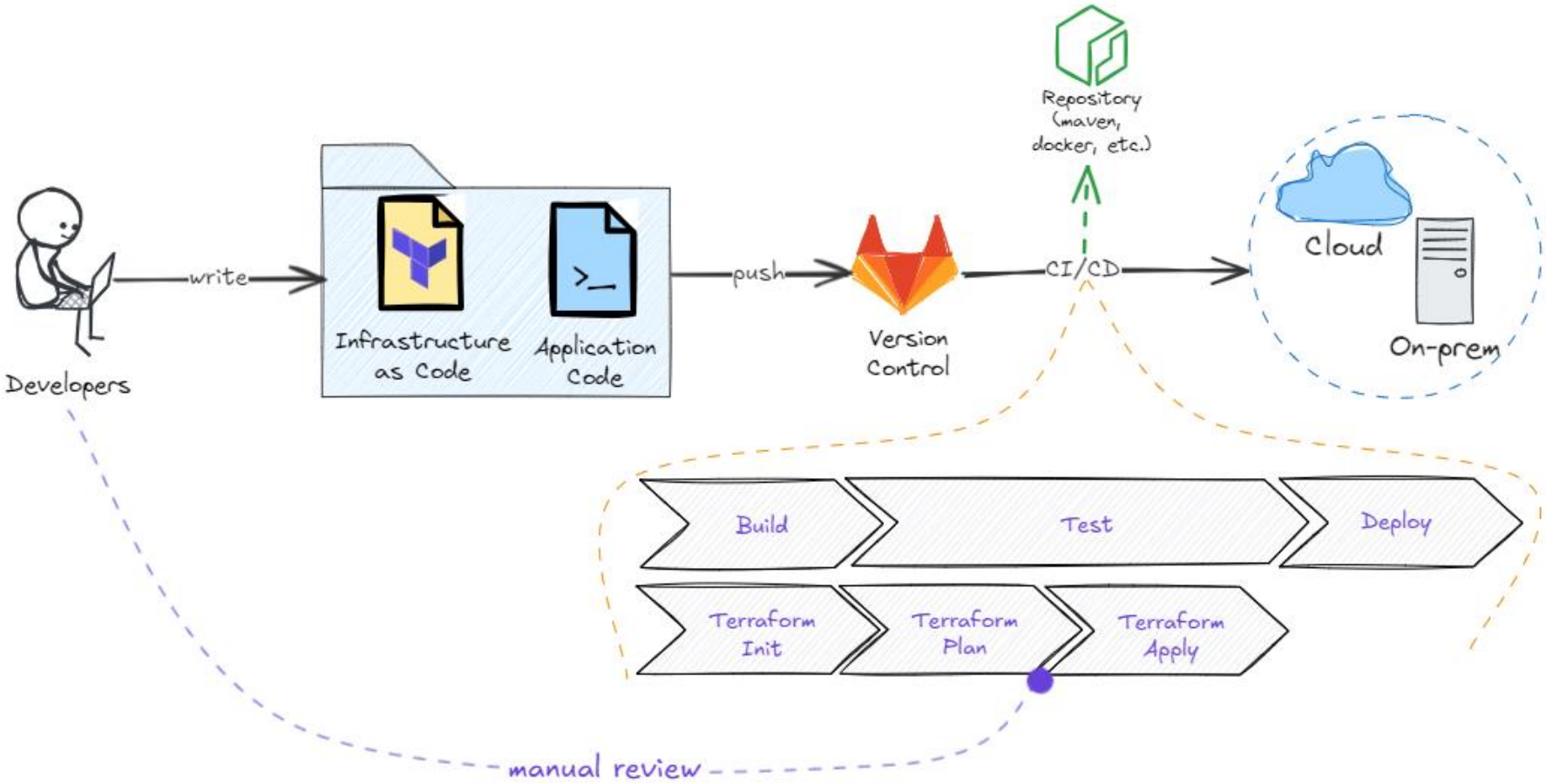


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IaC + GitOps

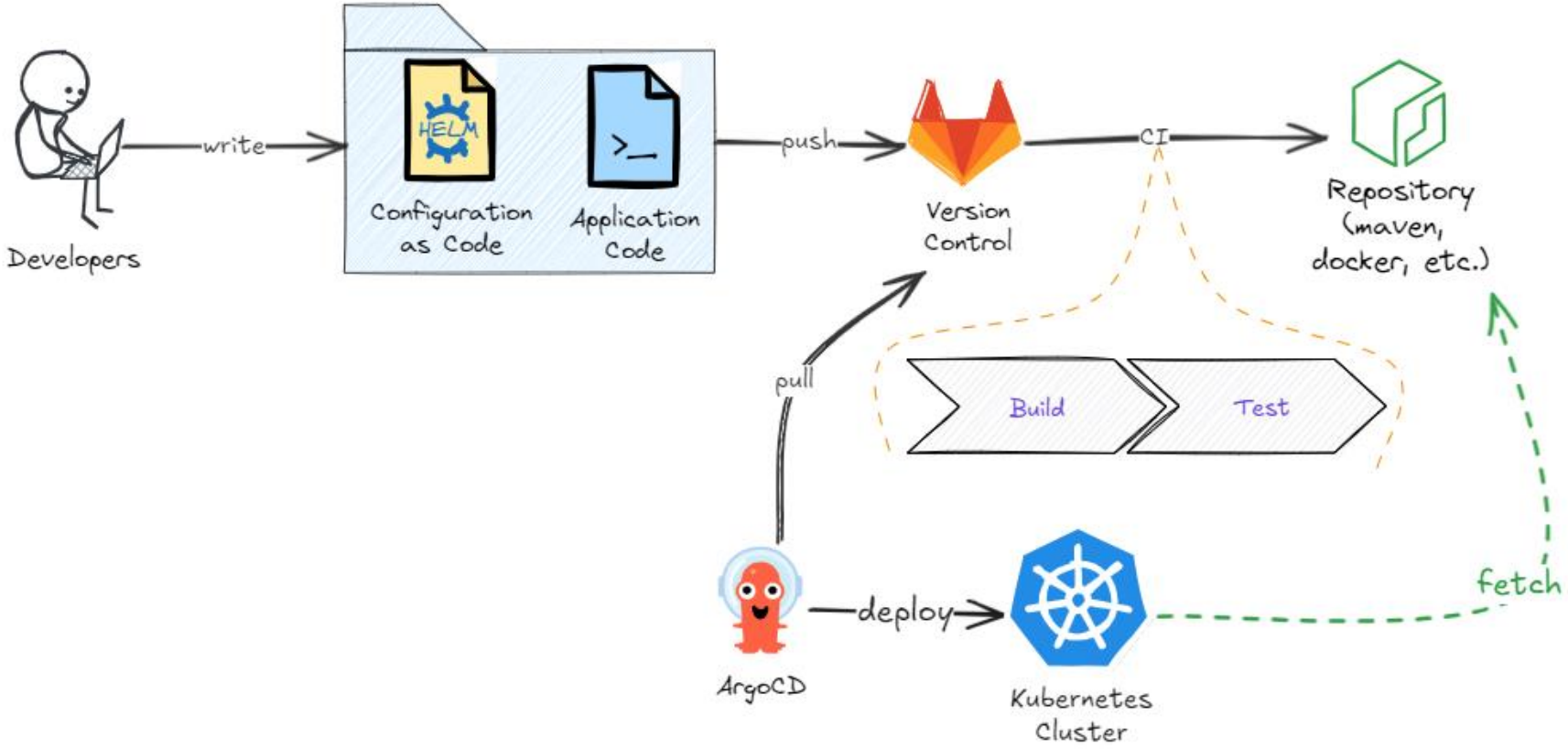
Terraform + Gitlab CI

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Kubernetes (Helm, ArgoCD)

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Q & A

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